Activity 5.1: Pythagorean Theorem & Special Triangles

A

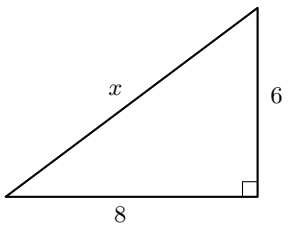
Name: _____ Partner's name: _____ Date: _____

Directions: Find the value asked for. Leave radicals in simplest form. Diagrams are accurate, but use math rather than a ruler or protractor.

Work alone, then compare after each problem. If answers differ, find the mistake together.

1

Find x.



Answer: _____ Match? ☐ Yes ☐ No

2

A right triangle has legs 5 and 12. Find the length of the hypotenuse.

Answer: _____ Match? ☐ Yes ☐ No

3

Is a triangle with sides 5, 7, and 9 acute, right, or obtuse?

Answer: _____ Match? ☐ Yes ☐ No

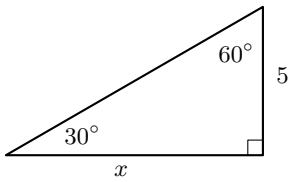
4

A 45°-45°-90° triangle has legs of length 7. Find the hypotenuse.

Answer: _____ Match? ☐ Yes ☐ No

5

Find x.



Answer: _____ Match? ☐ Yes ☐ No

6

A 45°-45°-90° triangle has hypotenuse 12. Find the length of each leg.

Answer: _____ Match? ☐ Yes ☐ No

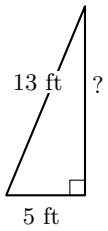
7

In a 30°-60°-90° triangle, the short leg is 8. Find the hypotenuse.

Answer: _____ Match? ☐ Yes ☐ No

8

A 13-ft ladder leans against a wall with its base 5 ft from the wall. How high up the wall does it reach?



Answer: _____ Match? ☐ Yes ☐ No

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B

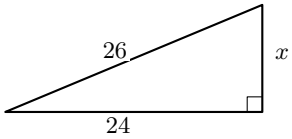
Name: _____ Partner's name: _____ Date: _____

Directions: Find the value asked for. Leave radicals in simplest form. Diagrams are accurate, but use math rather than a ruler or protractor.

Work alone, then compare after each problem. If answers differ, find the mistake together.

1

Find x.



Answer: _____ Match? ☐ Yes ☐ No

2

Find the distance from $(0, 0)$ to $(5, 12)$ on the coordinate plane.

Answer: _____ Match? ☐ Yes ☐ No

3

Is a triangle with sides 6, 8, and 11 acute, right, or obtuse?

Answer: _____ Match? ☐ Yes ☐ No

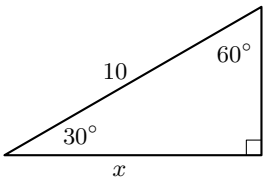
4

Find the length of the diagonal of a square with side length 7.

Answer: _____ Match? ☐ Yes ☐ No

5

Find x.



Answer: _____ Match? ☐ Yes ☐ No

6

A square has a diagonal of length 12. Find the side length of the square.

Answer: _____ Match? ☐ Yes ☐ No

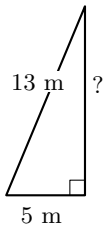
7

In a 30°-60°-90° triangle, the long leg is $8\sqrt{3}$. Find the hypotenuse.

Answer: _____ Match? ☐ Yes ☐ No

8

A 13-m guy wire runs from the top of a pole to the ground, 5 m from the pole's base. How tall is the pole?



Answer: _____ Match? ☐ Yes ☐ No